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Route to Victory

Abstract

This paper analyzes offensive passing performance in the 2023 NFL season using player-tracking data from the 2026 NFL Big Data Bowl. Three analyses were conducted: pre-snap prediction, route effectiveness, and ball-tracking efficiency. The pre-snap OLS model achieved an R^2 of 0.531, with pass completion status being the strongest predictor of yards gained. Play action and quarterback scrambles each increased expected yards by approximately 1 yard. Route effectiveness analysis across 396,796 targeted receiver observations revealed that Post, Corner, Go, and Wheel routes maximized yards gained, while Flat, Hitch, Out, and Cross routes achieved the highest completion rates. The Cross route emerged as the most balanced option overall. Ball-tracking efficiency (BTE) analysis showed that receivers run paths 23% longer than the direct distance, reflecting strategic route complexity. Player position, team schemes, and route type were the strongest BTE predictors.

Introduction

Player-tracking technology has rapidly expanded in the National Football League (NFL) in recent years. Football analytics have been transformed and are providing unprecedented information about where players are, how they move, and what is happening on the field throughout each play. The 2026 NFL Big Data Bowl provides structured datasets that allow analysts, students, and professionals to explore on-field behavior using modern statistical techniques such as ordinary least squares (OLS) and logistic regression. In this project, several datasets from the Big Data Bowl will be leveraged, including supplemental pre-snap information and route level input and output files. These will examine how measurable player and team

factors relate to offensive performance. We aim to translate situational and player tracking variables into interpretable insights based on empirical evidence.

This paper proceeds in three parts. First, the supplemental pre-snap dataset will model yards gained on any given play as a function of formation, alignment, coverage indicators, and other pre-snap variables. The pre-snap OLS model will focus on which factors help predict offensive success in the passing game. Next, route effectiveness will be explored by combining the input and supplementary datasets into one master set. The master set will be used to evaluate which receiver routes generate the most favorable outcomes and under what conditions. Finally, ball-tracking efficiency will be assessed. The master dataset and regression tools will be used to understand how player movement and ball placement correspond to play results. A 360-degree analysis will be produced, involving the main aspects of the offensive passing game. Pre-snap information, effective routes during the play, and reflecting on efficiency after the play capture all components of a football play. Together, these 3 sections will show how predictive analytics can reveal patterns in strategy, execution, and performance within the NFL's player tracking data.

Literature Review

Pro Football Focus (PFF) did a study on which route concepts create the most separation and the most open space. This analysis is helpful when arguing that receiver routes are significant predictors of yards gained. One of the main takeaways from this study was that the slant route was the only concept to rank top three in both open rate and average separation in the 2023 NFL season, as shown in Figure 1.

Figure 1

Highest Open Rate and Avg. Separation Grade by Route Concept | 2023 Season

Route Concept	Open Rate	Route Concept	Avg. Separation Grade
Slant	62.7%	Comeback	0.10
Whip (In-Breaking)	62.4%	Post	0.09
Wheel	61.0%	Slant	0.05

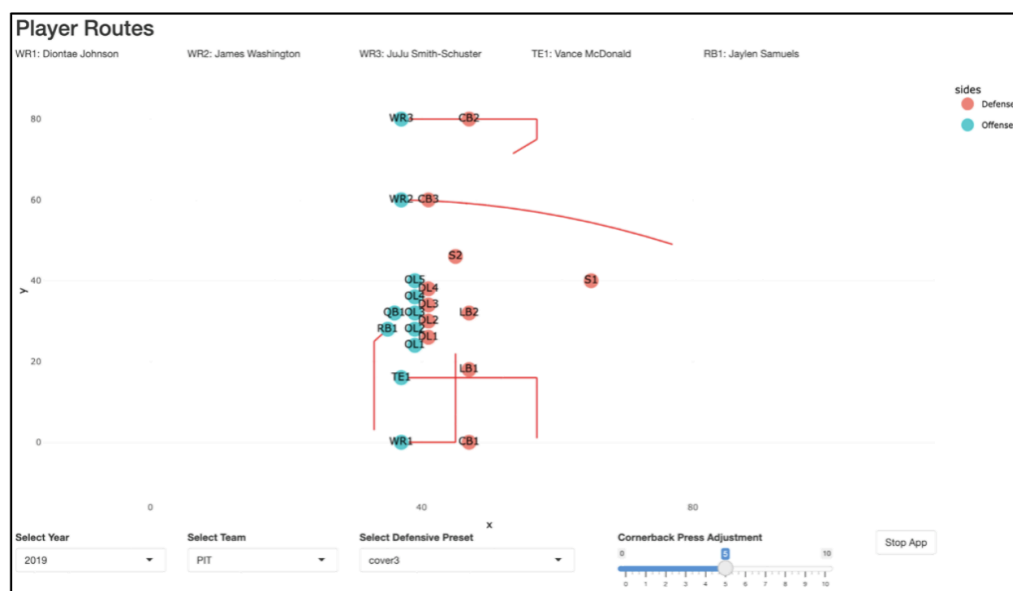
Figure 1, demonstrating the effectiveness of the slant route, foreshadows a key finding in our research. The article states additional factors that can be further investigated, such as receiver alignment, pre-snap motion, route depth, alignment, and coverage scheme. (Smith, 2024). The predictors listed are consistent with the analysis seen in this report about predicting passing yards gained.

Further research evaluates quarterback performance and how that translates to team success. After the first six weeks of the 2017 season, the six teams that had QBs with a passer rating of 100 or better led in terms of win percentage (Reyers and Swartz, 2022). This is suggestive that the NFL passer rating, which leads to passing yards gained, is a function of team success. Prediction has always been such a vital part of a football game that teams devote numerous hours and resources towards gaining insights into an opponent's tendencies. Assessing a team's propensity to run or pass the ball in a given situation is at the forefront of defensive coaching philosophies (Lee et al, 2016). Furthermore, determining pre-snap features that can lead to an accurate prediction is of the utmost importance, hence motivating the pre-snap analysis.

The first related literature to a route analysis comes from an undergraduate college student at the University of Arkansas, Fayetteville, who published a thesis based on "The

Quantitative Analysis and Visualization of NFL Passing Routes” (Chitturi, 2024). To fulfill the requirements for the visualization aspect of the thesis, a UI was made taking the following parameters: “the user inputted new choices as it pertains to the coverage type, year, team, or cornerback press adjuster” (Chiturri, 2024). Figure 2 shows an example of said UI, where an accurate representation of a “Sample visualization of routes (Pittsburgh Steelers under cover 3, 2019)” (Chitturri, 2024).

Figure 2



To continue to expand into literature related to routes and pass completion, Marc Schmid, Patrick Blauburger, and Martin Lames worked on a paper titled: “Simulating Defensive Trajectories in American Football for Predicting League Average Defensive Movements” (Schmid et al, 2021). When discussing the Pass Completion Probability Model, the authors concluded that “sophisticated models like this are very suitable for analyzing quarterback decision-making and even whole plays in-depth, but are too complex to assess team performance” (Schmid et al, 2021).

Ball tracking efficiency (BTE) is a metric used to evaluate the nature of a play or route by comparing the actual distance a player covers to reach the ball with the “perfect” straight line distance. Lower BTE scores do not necessarily indicate a less efficient route or a poor performance; rather, they can reflect the strategic complexity of the play. To the casual observer, a direct route to the ball may seem most efficient, but many football plays are intentionally designed to deceive the opposing team, even if this means players cover longer, less direct paths. The first article, “*Expertise and Deceptive Movements in Sport*” (Raffan et al, 2024), provides a framework for understanding deception in sports, distinguishing between deceit, in which an athlete conveys false information, and disguise, in which the true outcome of an action is concealed. In football, this highlights how seemingly inefficient routes may actually be purposeful, requiring defenders to detect and respond to deception effectively. Similarly, “*The Kinematics of False Intent Conveyed by Deceptive Sidestep Actions*” (Warren-Westgate et al, 2024) examines deceptive movements in rugby, which shares many similarities with football. The study found that players executing deceptive actions display certain kinematic cues, such as “exaggerated head rolls and center-of-mass displacements”, which may signal false intent. These cues are particularly relevant to BTE, as observers must process complex, misleading information to accurately anticipate a player’s path and intercept the ball.

Data

Overview

The data for our project comes from the 2026 NFL Big Data Bowl, an analytics competition that provides player tracking information from every passing play across the 2023 NFL season. The full dataset consists of 3 main components: Input data, which contains tracking

information before the pass is thrown, Output data, which contains tracking information after the throw is complete, and Supplementary data, which contains additional information about player level statistics and game specific information. All the datasets share common variables: `game_id` and `play_id`, which can be used to match the datasets if needed. Players are linked using `nfl_id`.

The input data contains frame by frame positional information for every player on the field, represented by x and y coordinates, speed (s), acceleration (a), orientation (o), and movement direction (dir). It also includes the player's height, weight, position, and role in the play, as well as the predicted ball landing point (`ball_land_x` and `ball_land_y`). The output data contains the player position (x, y) after the pass was thrown.

The supplementary data adds contextual information about each play, including: the season and week of the game, yards to go, offensive formation, receiver alignment, route of targeted receiver, and yards gained. These variables improve the accuracy of modeling situationally dependent plays, which will be explored in the analysis.

For this 3 part analysis of football performance, the datasets were merged, prepared, and cleaned to best fit the research questions and models. The following sections describe the data transformations for each individual analysis.

Pre-Snap Data

The pre-snap analysis will pull from a supplemental passing dataset from the 2023 NFL season. It has 41 columns with 18,009 rows. The continuous dependent variable is yards gained, which measures the amount of progress that the offensive team makes each play. There are many other predictors included that can predict yards gained, such as pass result, pass length, dropback type, defensive coverage, down, and distance. Figure 3 displays the summary statistics of yards gained from the supplemental data set.

Figure 3
yards_gained

count	18009.00
mean	8.15
std	9.97
min	-22.00
25%	0.00
50%	6.00
75%	12.00
max	92.00

Routes Data

The dataset classifies all routes into 12 distinct categories, as seen in Figure 4, where the most attempted routes are Hitch, Out, and Flat. Differently, the routes with the highest completion rates are Screen, Flat, and Angle. The route analysis will include different models, which will use either “Yards Gained” or “Pass Result” as their dependent variable. Both of these variables are meant to gauge how effective each route is, where “Pass Result” shows the outcome of said route, and “Yards Gained” showcases the different importance of completed passes by route. Furthermore, the route analysis only includes targeted receivers, meaning that only the observations where the receiver was actually thrown the ball to were kept.

Figure 4

Summary Statistics by Route		
Route	Attempts	Completion Rate
SCREEN	7,774	0.846
FLAT	55,688	0.835
ANGLE	17,091	0.828
HITCH	71,550	0.747
SLANT	21,712	0.703
CROSS	48,183	0.696
OUT	59,046	0.689
IN	34,283	0.594
POST	23,409	0.533
CORNER	16,219	0.507
WHEEL	2,203	0.460
GO	39,638	0.417
Note: Routes sorted by completion rate (highest to lowest).		

Ball Tracking Data

This analysis aimed to investigate which variables best predicted the nature of a route, and by how much the player had to deviate from the straight line distance to the ball. The variable Ball Tracking Efficiency (BTE) was created to represent the distance a player covered to reach the ball's landing coordinates, relative to the “perfect” straight line distance (see methodology section for details). All necessary information was obtained in the input and supplementary datasets.

Figure 5

```
Final valid BTE measurements: 2,625
BTE Summary Statistics:
count    2625.000000
mean     0.810823
std      0.161464
min      0.068642
25%      0.730673
50%      0.853726
75%      0.935067
max      0.999915
```

To improve accuracy, the analysis focused on players who were marked as the Targeted Receiver for a given play, with all observations where players were not targeted receivers removed. After removing the targeted receivers, the dataset had 396,914 observations; however, this number was drastically reduced when calculating one BTE per play and then averaging these values per player. The dependent variable for the analysis was each player's average BTE; the dataset contained 2,625 valid BTE measurements across 62 variables, with 310,528 training observations and 77,632 test observations. The average BTE for all players was 0.81, meaning most players ran routes 23% longer than the straight line distance, reflecting intentional strategic movements rather.

Methodology

Pre-Snap Analysis

Yards_gained will be the continuous dependent variable for the pre-snap analysis. Predicting this is valuable because it helps teams evaluate how defenses react to different offensive looks, alignments, and plays/routes that succeed. An ordinary least squares (OLS) model will be used to answer 2 questions. Which pre-snap variables matter the most when predicting yards gained? How does the OLS model perform?

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_n X_{ni} + \epsilon_i$$

- Y_i = yards gained (dependent variable),
- X_{1i} = independent variables capturing pre-snap features,
- $\beta_0, 1, 2, 3, \dots, n$ = estimated parameters representing the marginal effect of each predictor on yards gained,

- ϵ_i = error term.

It is hypothesized that a few major predictors will emerge from the offensive side of the ball, the defensive side, and other situational variables to predict the passing yards gained for an offense. These predictors consist of offensive formation, pass length, defensive coverage, down, and distance. Although a casual football fan has an expectation of which predictors can influence the yards gained on each play, this analysis is interested in the magnitude of coefficients from the OLS output model. Determining the magnitude of predictors that influence how many yards a team is expected to gain each play is crucial to coaches, players, and fans.

Route Analysis

For all the models produced by this analysis, the predictors are all the possible routes, which are transformed into dummy variables. For all the models, the worst-performing route will be dropped as a base value; therefore, all the route coefficients will be in reference to the dropped route. For the first segment of this analysis, “yards_gained” will be the dependent variable on an OLS regression model following the form:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_n X_{ni} + \epsilon_i$$

- Y_i = yards gained (dependent variable).
- $X_{1i}, 2, 3, \dots, n$ = routes dummies.
- β_0 = intercept.
- $\beta_1, 2, 3, \dots, n$ = estimated parameters representing the marginal effect of each predictor on yards gained.
- ϵ_i = error term.

The use of yards gained as a predictor is to measure the overall worth of each route. For instance, a Flat Route has a higher completion rate than a Corner Route, but the latter gains more yards

when completed, hence why “yards_gained” is an ideal dependent variable; it captures both the completion rate and route efficiency into one. For this OLS formula, four different models were generated (Model 1, 2, 3, and 4): Model 1 & Model 2 only included data from where all passes were completed, Model 2 had more control variables than Model 1; Model 3 & Model 4 were regressed on all the data available, including both completed and incompleted passes, and similarly to Model 2, Model 4 had more control variables compared to Model 3.

For the second segment of this analysis, “pass_complete” will be the dependent variable, using a Logistic regression model with the following formula:

$$\text{logit}(P(Y_i = 1)) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_n X_{ni}$$

- $P(Y_i = 1)$ = probability of passing completion.
- $\text{logit}()$ = log-odds transformation.
- $X_{1i}, X_{2i}, \dots, X_{ni}$ = routes dummies.
- β_0 = intercept.
- $\beta_1, 2, 3, \dots, n$ = coefficients for each predictor.

The aim of having “pass_complete” as the dependent variable is to purely focus on what routes have the highest success rates, which can be really important in situations where just moving forward is a priority. “Pass_complete” was slightly changed so that both an intercept and an incomplete pass counted as the same, simply to have a binary output. Using this Logistic regression formula, Model 5 & Model 6 were computed: Model 5 has fewer control variables than Model 6. The distribution of the control variables can be seen below in the table represented by Figure 6.

Figure 6

Control Variables <i>Model 1, Model 3, Model 5</i>	Control Variables <i>Model 2, Model 4, Model 6</i>
Coverage, position, down, yards to go, and absolute yardline number	Coverage, position, down, yards to go, absolute yardline number, formation, alignment, dropback, pass loc, speed, acceleration, dropback distance, defenders in the box, quarter, week, score differential, and play action

Lastly, both ridge and lasso models were run, but since neither offered any real improvement, they will not be shown in the results section.

Ball Tracking Efficiency Analysis

Ball Tracking Efficiency (BTE) can be defined as:

$$\text{Ball Tracking Efficiency} = \text{Ideal Distance} / \text{Actual Distance Ran}$$

Where:

Ideal Distance: Straight line from start position to ball landing

Actual Distance Ran: Sum of frame to frame distances traveled

A BTE score was calculated for each play in which a player was the targeted receiver. These values were then averaged for each player in the 2023 NFL season. By definition, BTE scores range from 0 to 1, where 1 represents a “perfect” straight line path to the ball’s landing coordinates. Figures 7 and 8 show the differences in a high BTE and low BTE play.

Figure 7 High BTE Example

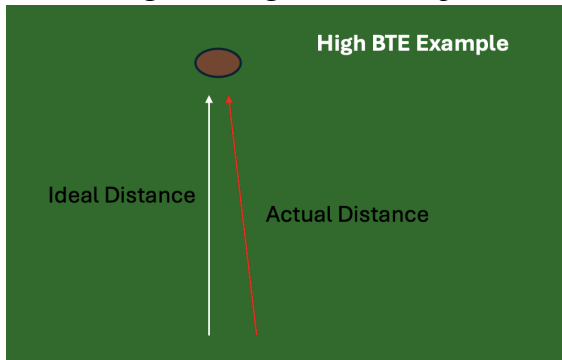
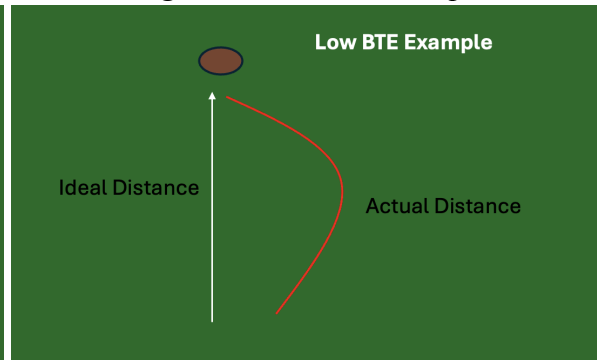


Figure 8 Low BTE Example



It is important to note that BTE is not a measure of performance or skill; it is simply a metric of route type. Players with lower BTE scores tend to run routes that curve more or have sudden, sharp “cuts” in them, whereas players with higher BTE scores tend to take routes that follow more direct paths. This metric can also be influenced by the design of the play, as some plays intentionally perform short passes, naturally producing direct routes. While others require longer or more complex routes, which do not indicate poorer performance.

This analysis aims to answer the following question: Which variables best predict a player’s BTE for a given route? Because the dependent variable is continuous (ranging from 0 to 1), 3 models have been used. Ordinary Least Squares (OLS), Ridge, and Lasso, each offering their unique strengths and weaknesses.

Results

Pre-Snap Analysis

Pass result is by far the strongest predictor, which intuitively makes sense. Pass results have 3 possible values: complete, incomplete, or interception. From the OLS output table in Figure 9, an incomplete pass results in the model predicting roughly 15 fewer yards gained compared to a complete pass. If the pass is intercepted, the model expects 17 fewer yards to be gained on the play on average. Pass result has a huge impact and is the strongest predictor when determining how many yards were gained. This makes sense because the only way to gain yards using passing data is to complete the pass. An incomplete pass results in 0 yards, and an interception means the defense caught the pass and now possesses the football.

A subtle detail that was not captured in the OLS output, but is still relevant to in-game decision making, is that play action helps. Play action means that the offensive team fakes a run play and then decides to pass the ball. A significant coefficient of 0.92 suggests that running play action increases the expected yards gained by about 1 yard. This aligns with analytics because play action tends to deceive the defenders and create more space for open receivers to catch the ball.

Dropback type is another useful predictor. The type of drop back by the quarterback (QB) helps predict success. The most common drop back is traditional, where the QB takes a few steps backward to throw. There are also designed plays where the QB can run to his left or right to throw. There are also times where the QB does not stick to the play and decides to “scramble,” that is, run to the left or right without it being called. The scramble dropback type was the most useful in terms of gaining more yards. On average, when the QB scrambles, expected yards gained increase by 1.6 yards compared to a designed rollout left (dropback_type baseline).

Scramble plays often result in broken coverages by the defense or QB runs that pick up yards.

Scrambles are unplanned and plays where the QB relies on his instincts. The output confirms that these types of dropbacks give the defense the most trouble because they happen less often and are difficult to prepare for.

The route that the receiver runs is very insightful in predicting the amount of yards gained on a given pass play. The OLS output shows that the wheel route provides the largest positive boost, with a coefficient of nearly 4 yards more than the baseline angle route. The post, corner, and go route are also impactful in a positive way relative to the angle route. Slants and in-routes had coefficients a little higher than 1 yard. The interpretation from the route predictor is that vertical and deep-breaking routes (go, post, corner, wheel) tend to generate significantly more yards, but are not completed at a relatively high rate. The slant and in-breaking routes are more reliable and completed at a higher rate. If the coach needs a short gain and a high chance of completion, slant and other short routes may be an optimal choice. If the coach is looking to take a bigger chance for more yards in a singular play, they could call a post or go route, for example.

The defenders in the box variable offers a slightly counterintuitive interpretation with a coefficient of +0.20 yards. The initial interpretation is that more defenders in the box leads to more yards, which does not seem to make sense. Theoretically, the more defenders that are present to stop the offense, should decrease predicted yards gained. However, knowing that the dataset only incorporates passing plays, this estimate makes sense. More defenders in the box means that there are less defenders to guard against long pass plays, thus giving up more yards than the baseline for each additional defender in the box. Another interesting feature is that the second down variable has a negative average yards gained coefficient, whereas the third down

estimate is positive. This suggests that third down features more aggressive playcalling, hence more yards gained on average.

The OLS results also demonstrated that formation and alignment matter less than quarterback decision, the type of route, coverage, and pass result. This is shown by many receiver alignment variables and offensive formation variables being deemed insignificant predictors. The key takeaways from the pre-snap regression are that completion status is the biggest yardage driver. Route type, dropback type, and whether the QB scrambles or not also determine if the offensive team will have more success through the air. It can also be concluded that most formations and alignments do not meaningfully impact yards gained as much as the other included variables do. The OLS model performs relatively well. The R^2 value equals 0.531. This means that roughly 53% of the variance in passing yards gained is explained by pre-snap variables from the supplemental dataset. This is a very solid model for football data.

Figure 9

<i>Dependent variable: yards_gained</i>	
OLS Model	
Intercept	3.680 (2.049) [-0.336, 7.696]
offense_formation[T.I_FORM]	1.353 (1.563) [-1.710, 4.417]
offense_formation[T.JUMBO]	-1.607 (1.894) [-5.320, 2.106]
offense_formation[T.PISTOL]	1.052 (1.511) [-1.910, 4.013]
offense_formation[T.WILDCAT]	-1.387 (4.000) [-9.227, 6.453]
pass_result[T.I]	-15.023*** (0.120) [-15.259, -14.788]
pass_result[T.IN]	-17.201*** (0.340) [-17.868, -16.533]
receiver_alignment[T.3x0]	-5.507 (6.951) [-19.132, 8.119]
receiver_alignment[T.3x2]	2.094 (1.925) [-1.679, 5.867]
receiver_alignment[T.3x3]	1.303 (5.209) [-8.906, 11.513]
receiver_alignment[T.4x0]	-2.223 (3.056) [-8.213, 3.767]
receiver_alignment[T.4x1]	2.343 (1.971) [-1.521, 6.207]
team_coverage_type[T.PREVENT]	1.932 (1.050) [-0.126, 3.991]
dropback_type[T.SCRAMBLE]	1.621*** (0.429) [0.779, 2.462]
route_CORNER	1.624*** (0.404) [0.832, 2.415]
route_GO	1.339*** (0.356) [0.641, 2.036]
route_IN	1.306*** (0.326) [0.667, 1.945]
route_POST	2.460*** (0.372) [1.730, 3.189]
route_SCREEN	-1.338** (0.445) [-2.210, -0.466]
route_SLANT	1.091*** (0.333) [0.438, 1.744]
route_WHEEL	3.853*** (0.755) [2.374, 5.332]
Observations	18,001
R ²	0.531
Adjusted R ²	0.530
F-statistic	398.9***
Note: Standard errors in parentheses. 95% confidence intervals in brackets. * p<0.1; ** p<0.05; *** p<0.01	

Route Analysis

OLS Regression

In this regression, Screen Route has been dropped as a base reference, as it performed the poorest in the following regression models. As stated in the method section above, this route analysis of OLS Models using “yards_gained” as the dependent variable features 4 different models, as seen in Figure 10 below. Model 1 & Model 2 have only 215,004 observations, compared to the 317,436 of Model 3 & Model 4, since they only include completed passes. Model 2 & Model 4 have more control variables, meaning that the models are more accurate in assessing the real values of the route coefficients. This four-model setup serves as a way to gradually show the change in coefficients from model to model, finishing at the most accurate model: Model 4, which has the most observations and the most control variables. An instance of this can be seen by the gradual diminishing change in the coefficient of route_post, where the coefficient performs the following journey across models, starting from Model 1: 19.523 → 18.434 → 8.994 → 8.377.

After establishing that Model 4 is the most accurate, it can be seen in Figure 10 that all route predictors are highly statistically significant, given that their p-values are below 0.01. Model 4 reveals that routes Post, Corner, Go, and Wheel perform the best in terms of “yards_gained,” with their respective coefficients being 8.377, 6.977, 6.894, and 6.129. A way to interpret these coefficients would be that the Post route gains, on average, 8.377 more yards compared to the Screen route, while keeping the other predictors constant. The bottom 3 performing routes, alongside the Screen route, are Hitch, Out, and Flat routes, with the following corresponding coefficients: 2.874, 1.999, 1.08.

To assess the overall performance of Model 4, the R^2 can be taken into consideration, which is the same as the adjusted R^2 for this model. The R^2 for Model 4 is 0.068, meaning that only roughly 6.8% of the variation in “yards_gained” is explained by the model, but this is explained by the fact that the model includes all pass attempts, where incompletions always result in zero yards regardless of route type or other predictors: as a comparison, the R^2 in Model 2 is 0.385. The F-statistic is 405.183 and highly statistically significant, meaning that Model 4 has a real predictive and explanatory power in determining some of the variation in “yards_gained.”

Figure 10

Route Effectiveness Analysis - Model Comparison				
	Dependent variable: yards_gained			
	Model 1 Completed Passes	Model 2 Completed Passes	Model 3 All Attempts	Model 4 All Attempts
const	5.074*** (0.516) [4.062, 6.085]	1.086 (0.821) [-0.524, 2.696]	4.805*** (0.526) [3.774, 5.836]	2.166** (0.855) [0.490, 3.842]
route_POST	19.523*** (0.151) [19.227, 19.819]	18.434*** (0.149) [18.142, 18.726]	8.994*** (0.154) [8.692, 9.295]	8.377*** (0.155) [8.073, 8.681]
route_CORNER	18.226*** (0.164) [17.905, 18.547]	16.489*** (0.162) [16.171, 16.807]	7.798*** (0.162) [7.480, 8.115]	6.977*** (0.163) [6.657, 7.297]
route_GO	22.463*** (0.145) [22.179, 22.748]	21.488*** (0.142) [21.210, 21.766]	7.340*** (0.146) [7.054, 7.626]	6.894*** (0.146) [6.608, 7.180]
route_WHEEL	18.349*** (0.333) [17.697, 19.002]	16.991*** (0.326) [16.352, 17.630]	6.886*** (0.283) [6.332, 7.440]	6.129*** (0.284) [5.572, 6.686]
route_CROSS	7.147*** (0.134) [6.885, 7.410]	5.499*** (0.133) [5.238, 5.760]	4.607*** (0.144) [4.325, 4.890]	3.789*** (0.145) [3.505, 4.073]
route_IN	8.333*** (0.141) [8.056, 8.609]	7.333*** (0.139) [7.060, 7.606]	4.040*** (0.148) [3.750, 4.330]	3.561*** (0.148) [3.271, 3.851]
route_ANGLE	4.319*** (0.153) [4.019, 4.619]	3.955*** (0.150) [3.661, 4.249]	3.437*** (0.166) [3.112, 3.762]	3.176*** (0.166) [2.851, 3.501]
route_SLANT	4.903*** (0.147) [4.615, 5.191]	6.107*** (0.144) [5.825, 6.389]	3.337*** (0.156) [3.031, 3.642]	3.791*** (0.156) [3.485, 4.097]
route_HITCH	4.330*** (0.130) [4.075, 4.585]	4.134*** (0.127) [3.885, 4.383]	2.994*** (0.140) [2.720, 3.268]	2.874*** (0.140) [2.600, 3.148]
route_OUT	3.962*** (0.131) [3.705, 4.219]	3.438*** (0.129) [3.185, 3.691]	2.284*** (0.141) [2.008, 2.561]	1.999*** (0.141) [1.723, 2.275]
route_FLAT	1.906*** (0.131) [1.650, 2.163]	1.173*** (0.128) [0.922, 1.424]	1.504*** (0.142) [1.226, 1.782]	1.080*** (0.142) [0.802, 1.358]
Observations	215,004	215,004	317,436	317,436
Control Variables	17	47	17	47
R^2	0.349	0.385	0.058	0.068
Adjusted R^2	0.349	0.385	0.057	0.068
Residual Std. Error	8.799 (df=214975)	8.551 (df=214947)	10.430 (df=317407)	10.374 (df=317378)
F Statistic	4109.107*** (df=28; 214975)	2402.206*** (df=56; 214947)	692.450*** (df=28; 317407)	405.183*** (df=57; 317378)

Note: Standard errors in parentheses. 95% confidence intervals in brackets.
*p<0.1; **p<0.05; ***p<0.01

Logistic regression

In the following models for this Logistic regression, the Go route was dropped as a base reference since it performed the worst in this model. As stated in the methods section, “pass_complete” is the dependent variable, and Models 5 & 6 aim to figure out which routes have the best completion rate, regardless of how many yards they gain. Exactly like it was done with the OLS models, Model 6 has more control variables than Model 5, making it the most accurate model. As seen in Figure 11, almost all routes were statistically significant (p-value < 0.05) with the sole exception of the Wheel route, which had no statistical significance. This just means that the Wheel route and Go route are almost identical when it comes to pass completion. The best performing routes in pass completions were: Flat, Hitch, Out, and Cross, with respective coefficients of 0.621, 0.495, 0.373, and 0.395. The correct way to interpret one of these coefficients would be the following: the Flat route has 0.621 higher log odds of completion in comparison to the Go route, while keeping all the other predictors constant. On the flipside, the worst-performing routes alongside GO and Wheel were Post and Corner, with respective coefficients of 0.099 and 0.084. Model 6 has a 69.9 % accuracy in predicting pass completion, demonstrating that route selection and game situations are meaningful predictors of completion probability.

Figure 11

Route Completion Probability - Logistic Regression Results

	<i>Dependent variable: pass_complete</i>	
	Model 5	Model 6
const	0.814***	0.823***
route_FLAT	0.630** (0.008) [0.615, 0.644]	0.621** (0.007) [0.608, 0.636]
route_HITCH	0.513** (0.006) [0.502, 0.526]	0.495** (0.006) [0.485, 0.509]
route_OUT	0.388** (0.005) [0.376, 0.397]	0.373** (0.006) [0.361, 0.384]
route_CROSS	0.376** (0.005) [0.365, 0.386]	0.395** (0.005) [0.385, 0.406]
route_ANGLE	0.357** (0.006) [0.346, 0.370]	0.342** (0.006) [0.332, 0.356]
route_SLANT	0.285** (0.005) [0.275, 0.294]	0.240** (0.005) [0.232, 0.251]
route_SCREEN	0.258** (0.005) [0.249, 0.266]	0.250** (0.005) [0.241, 0.258]
route_IN	0.195** (0.005) [0.186, 0.204]	0.196** (0.005) [0.188, 0.206]
route_POST	0.096** (0.004) [0.088, 0.102]	0.099** (0.004) [0.090, 0.106]
route_CORNER	0.071** (0.004) [0.063, 0.078]	0.084** (0.005) [0.076, 0.092]
route_WHEEL	0.008* (0.004) [0.001, 0.016]	0.004 (0.004) [-0.003, 0.012]
Observations	317,436	317,436
Control Variables	17	47
Accuracy	0.696	0.699
AUC	0.679	0.690

Note: Standard errors in parentheses. 95% confidence intervals in brackets.
*p<0.1; **p<0.05; ***p<0.01

Figure 12 portrays a combination of both OLS and Logistic Models ranking routes based on their performance in each model: Rank 1 means that the route was the best performing in that model, and Rank 12 means that the route was the worst performing in that model. This is meant to assess both how many yards a route could potentially gain and the likelihood of said route being completed. Figure 12 shows that the clear winner is the Cross route with a total rank of 8, being in the top 5 performing routes in both OLS and Logistic regression; alongside the Cross

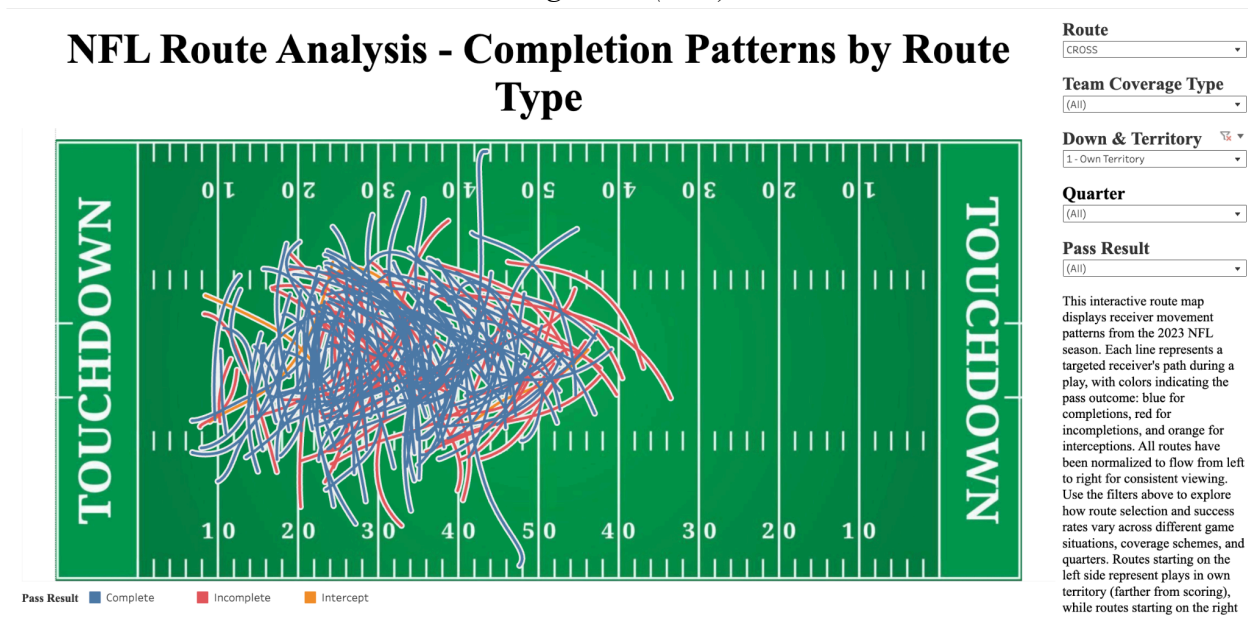
route, Post and Hitch routes are also good viable options. At the bottom of the table, Go, Wheel, and Screen routes are present, meaning that they perform worse than many of the other routes in most situations; therefore, this means that these routes should only be used in seldom specific situations.

Figure 12

Route	OLS Rank	Logistic Rank	Total Rank
Cross	5	3	8
Post	1	9	10
Hitch	9	2	11
Corner	2	10	12
Flat	11	1	12
Angle	7	5	13
In	6	8	14
Slant	8	6	14
Out	10	4	14
Go	3	12	15
Wheel	4	12	16
Screen	12	7	19

To finish this route analysis, Figure 13 shows a screenshot of an interactive Tableau visualisation, which is meant to be used as a means of independent analysis. The visualisation lets the user filter by route, team coverage type, down & territory, quarter, and pass results; as an example, Figure 13 shows all the Cross routes against all team coverages in the 1st down in the own territory, across all quarters.

Figure 13 ([Link](#))



Ball Tracking Efficiency Analysis

A train-test split was implemented to evaluate how each model would perform on unseen data. The OLS model had a test R^2 of 0.1943 and a test RMSE of 0.0656, explaining approximately 19.8% of the variance in BTE. The ridge model ($\lambda = \sim 1.15$) had a test R^2 of 0.1943 and a test RMSE of 0.0656, indicating that the predictors exhibit very minimal multicollinearity. Lastly, the lasso model ($\lambda = \sim 0.001$) had a test R^2 of 0.1720 and a test RMSE of 0.665, explaining slightly less of the variation in BTE compared to OLS and Ridge. Figure 15 shows the coefficients of the OLS Model.

The OLS model was selected as the best model because, although having the same test R^2 and RMSE, OLS achieves the same predictive accuracy without the additional complexity of the ridge model. The following interpretation focuses on the statistically significant coefficients from the OLS model.

Figure 14

MODEL COMPARISON SUMMARY				
Model	Train R^2	Test R^2	Train RMSE	Test RMSE
OLS	0.198319	0.194258	0.064848	0.065572
Ridge	0.198319	0.194258	0.064848	0.065572
Lasso	0.175186	0.171973	0.065777	0.066473
✓ Best model by Test R^2 : OLS ($R^2 = 0.1943$)				

Overall, the strongest predictors of BTE can be categorized into 3 main groups: player position, team and route, along with a few additional variables, outside of these categories.

Player position became one of the strongest predictors of BTE. Positions such as Fullback (FB) and Tight End (TE) had the largest negative coefficients, -0.254 and -0.183, respectively, when it came to predicting BTE, compared to the base case, Center Back (CB). CBs rarely catch the ball, but when they do, it is usually along a direct route, resulting in higher BTE. In contrast, FBs and TEs often run routes with more deception or lateral movement to get open, naturally reducing their BTE. The Tackle (T) position had a small negative coefficient of -0.060, reflecting that they rarely catch the ball, and when involved, it is usually in special plays with more direct paths, thus their BTE is much closer to 1.

The team in possession was another significant predictor. Coefficients ranged from around -0.090 (HOU) to +0.026 (ATL, LV) compared to the base case team ARI. Teams with

complex, movement intensive schemes (HOU, SF), -0.090 and -0.068, respectively, tended to have lower BTE, consistent with their use of misdirection and indirect routes. Teams emphasizing more vertical or straightforward passing (ATL, LV), 0.026 and 0.026, respectively, showed higher BTE, indicating straighter routes toward the ball. This systematic variation across teams supports BTE as a meaningful metric for capturing differences in offensive play styles.

Lastly, route types also play an important role in determining BTE. SCREEN passes exhibited the lowest BTE of -0.027, which reflects their backward movement nature. POST routes showed the highest BTE of 0.006, consistent with their typically straight, vertical trajectories. This perfectly aligns with real-life football, where routes with short, direct passes have higher BTEs, while ones that intentionally try to fool the opposing team by going one way then another have much lower BTEs. The base case for the route category was ANGLE.

Some predictors that didn't fit into the main categories also influenced BTE. Offensive formation WILDCAT, which is a trick formation where the ball is "snapped" to a running back instead of a quarterback, had a positive coefficient of 0.0462, compared to the baseline EMPTY offensive formation. This makes sense, as the receivers in this formation often run straighter routes, increasing BTE. Receiver alignment 3x0, which means 3 receivers line up on one side of the field and none on the other, had a coefficient of 0.038, suggesting that receivers have more space on that side of the field and can take more direct routes, relative to the baseline 1x1 alignment.

Figure 15

Ball Tracking Efficiency Model

<i>Dependent variable: avg_bte</i>	
	Model
player_position_FB	-0.254*** (0.004) [-0.261, -0.247]
player_position_T	-0.060*** (0.012) [-0.083, -0.038]
player_position_TE	-0.183*** (0.003) [-0.189, -0.177]
possession_team_ATL	0.026*** (0.001) [0.024, 0.028]
possession_team_HOU	-0.090*** (0.001) [-0.091, -0.088]
possession_team_LV	0.026*** (0.001) [0.024, 0.028]
possession_team_SF	-0.068*** (0.001) [-0.070, -0.066]
offense_formation_WILDCAT	0.046*** (0.007) [0.032, 0.060]
receiver_alignment_3x0	0.038** (0.015) [0.009, 0.067]
route_CORNER	0.003*** (0.001) [0.001, 0.005]
route_POST	0.006*** (0.001) [0.004, 0.007]
route_SCREEN	-0.027*** (0.001) [-0.029, -0.024]
pass_location_type_UNKNOWN	0.034*** (0.008) [0.018, 0.050]
Observations	310,528
Control Variables	126
R ²	0.198
Adjusted R ²	0.198
F Statistic	609.4*** (df=126; 310401)

Note:

Standard errors in parentheses. 95% confidence intervals in brackets.

*p<0.1; **p<0.05; ***p<0.01

Conclusion

This 3-way analysis provided an in-depth explanation of key factors and drivers in American Football. From the pre-snap analysis, the OLS model clearly showed that the pass result is the strongest predictor for yards gained, as if the pass isn't completed, 0 yards are gained. Other meaningful predictors include play action, increasing yards gained on average by about 1; quarterback scrambles, increasing average yards gained by roughly 1.6 in comparison to the dropback_type baseline (designed rollout left); and wheel routes, which increase yards gained by an average of roughly 4 yards in comparison to the angle baseline route. Overall, the model performed relatively well, with an R^2 of 0.531, meaning that roughly 53% of the variance in passing yards is explained by pre-snap variables, creating a very solid model for football data.

Regarding the route analysis, OLS and Logistic regression models were created as a means to compare different routes' impact on the continuous variable "yards_gained" and the categorical variable "pass result." The OLS Model clearly portrayed that the Post, Corner, Go, and Wheel routes were the best performing routes with an average of yards gained of at least 6, compared to the Screen baseline route, while keeping the other predictors constant. The Logistic Model highlighted the Flat, Hitch, Out, and Cross routes as the best performing routes in terms of pass completion, with an average of log-odds of at least 0.37 higher than the Go baseline route. Finally, using a ranking system to combine the two models, the podium of best performing routes is the following: 3rd place - Hitch route; 2nd place - Post route; and 1st place Cross route. This means that the Cross route is the most optimal when taking into account yards gained and pass completion.

Lastly, the Ball Tracking Efficiency analysis focused on an OLS regression model with "Ball Tracking Efficiency (BTE)" as its dependent variable, where BTE = 1 refers to a player

taking the perfect straight line to the ball. The model showed an average of 0.81 BTE, meaning that on average, a receiver runs 23% more than the “perfect” straight line distance. An important distinction to be made is that a lower BTE does not necessarily mean a worse player/route/performance; it actually indicates route and play complexity or a team play style. For instance, TE/WR/RB have lower BTE because those positions typically run more complex routes. Another finding is that Plays like Screen Passes or Flat Routes are typically lateral, resulting in lower BTEs, while plays like Post Routes have higher BTEs because of their more straight downfield nature. Overall, the predictor with the highest impact was the offensive formation wildcat, with a coefficient of 0.0462, meaning that this formation will have, on average, a higher BTE score by 0.0462 when compared to the base case offensive formation of empty. To conclude, BTE is not a measure of a player’s performance; rather, it measures route geometry for understanding offensive philosophy and efficiency.

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